

CHAPTER 5: SYSTEM DESIGN

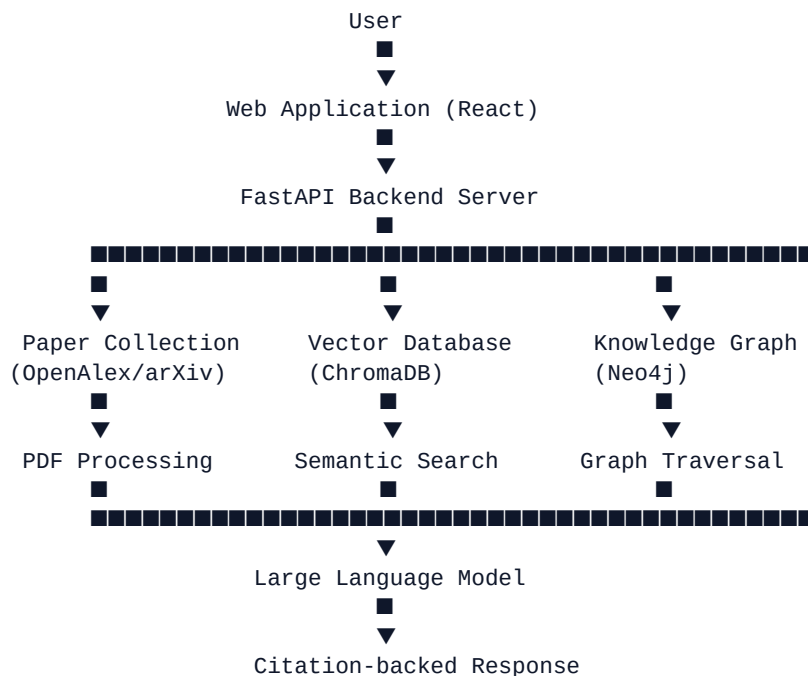
System Architecture, Data Flow Diagrams, UML & Component Specifications

5.1 Introduction

System design is a crucial stage in software development that defines the architecture, components, data flow, and interactions within the proposed system. A well-designed architecture improves scalability, maintainability, and performance while simplifying future enhancements.

The proposed ResearchGraph AI platform follows a modular architecture that separates data collection, document processing, knowledge extraction, semantic retrieval, and response generation into independent components.

5.2 Overall System Architecture

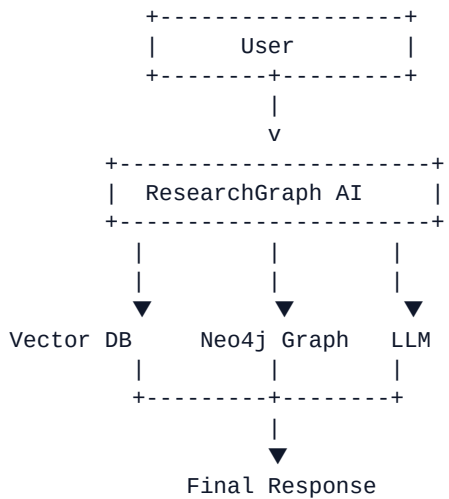


5.3 Architectural Description

- **Presentation Layer:** Web interface built with React, TypeScript, and Tailwind CSS. Provides dashboards for Search, AI Chat, Literature Review, Comparison, and Graph Visualization.
- **Application Layer:** FastAPI backend controller managing business logic, routing, AI model integration, and DB queries asynchronously.
- **Data Collection Layer:** Ingests metadata and PDFs from OpenAlex and arXiv APIs.

- **Document Processing Layer:** Parses PDFs using PyMuPDF and splits text into context-preserving chunks using LangChain.
- **Semantic Retrieval Layer:** BAAI/BGE-large-en-v1.5 embeddings indexed in ChromaDB for dense vector similarity search.
- **Knowledge Graph Layer:** Neo4j property graph mapping papers, authors, datasets, methods, and algorithms with rich relationships.
- **AI Response Generation Layer:** LangChain combines vector chunks and graph contexts to feed the LLM for citation-backed output.

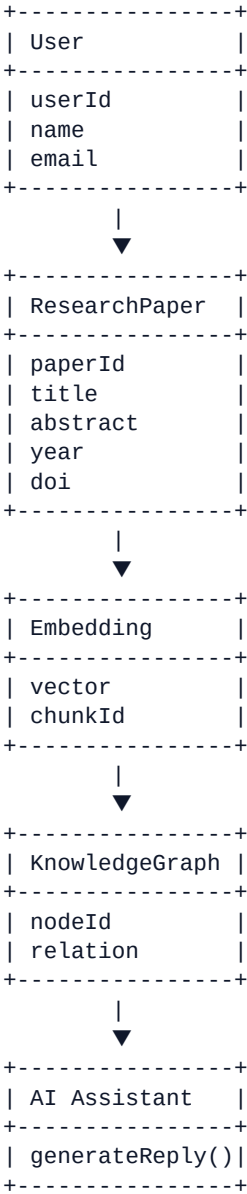
5.4 Data Flow Diagram (Level 0)



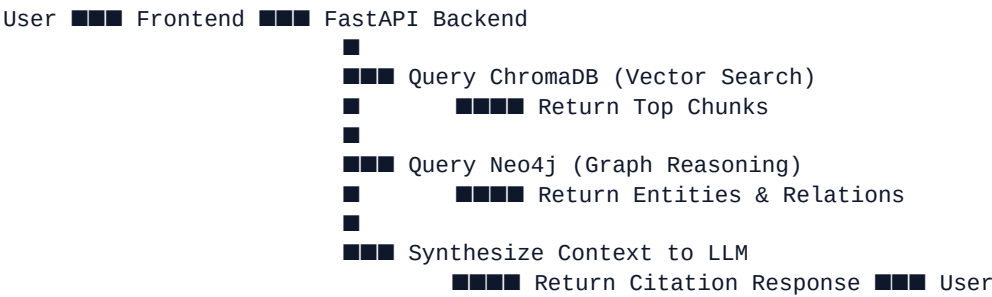
5.5 Data Flow Diagram (Level 1)



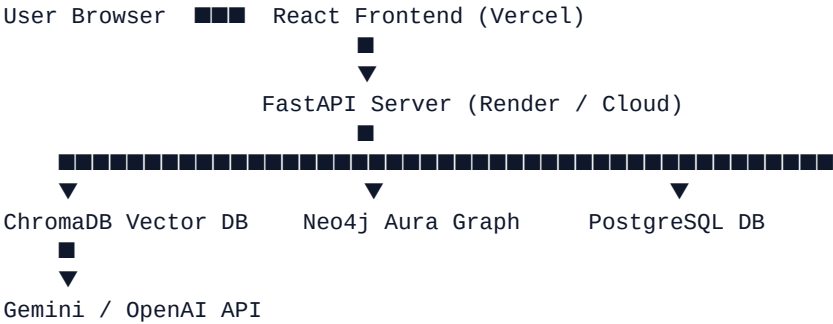
5.6 Class Diagram



5.7 Sequence Diagram & Activity Flow



5.8 Component & Deployment Diagrams



5.9 Design Advantages

- **Modular Architecture:** Allows independent updates to frontend components, databases, and AI models without breaking system integrity.
- **Explainable GraphRAG:** Combines semantic vector similarity with explicit graph relationships for verifiable AI reasoning.
- **Extensible Integration:** Effortlessly supports additional paper repositories (IEEE, Springer) or newer LLMs.